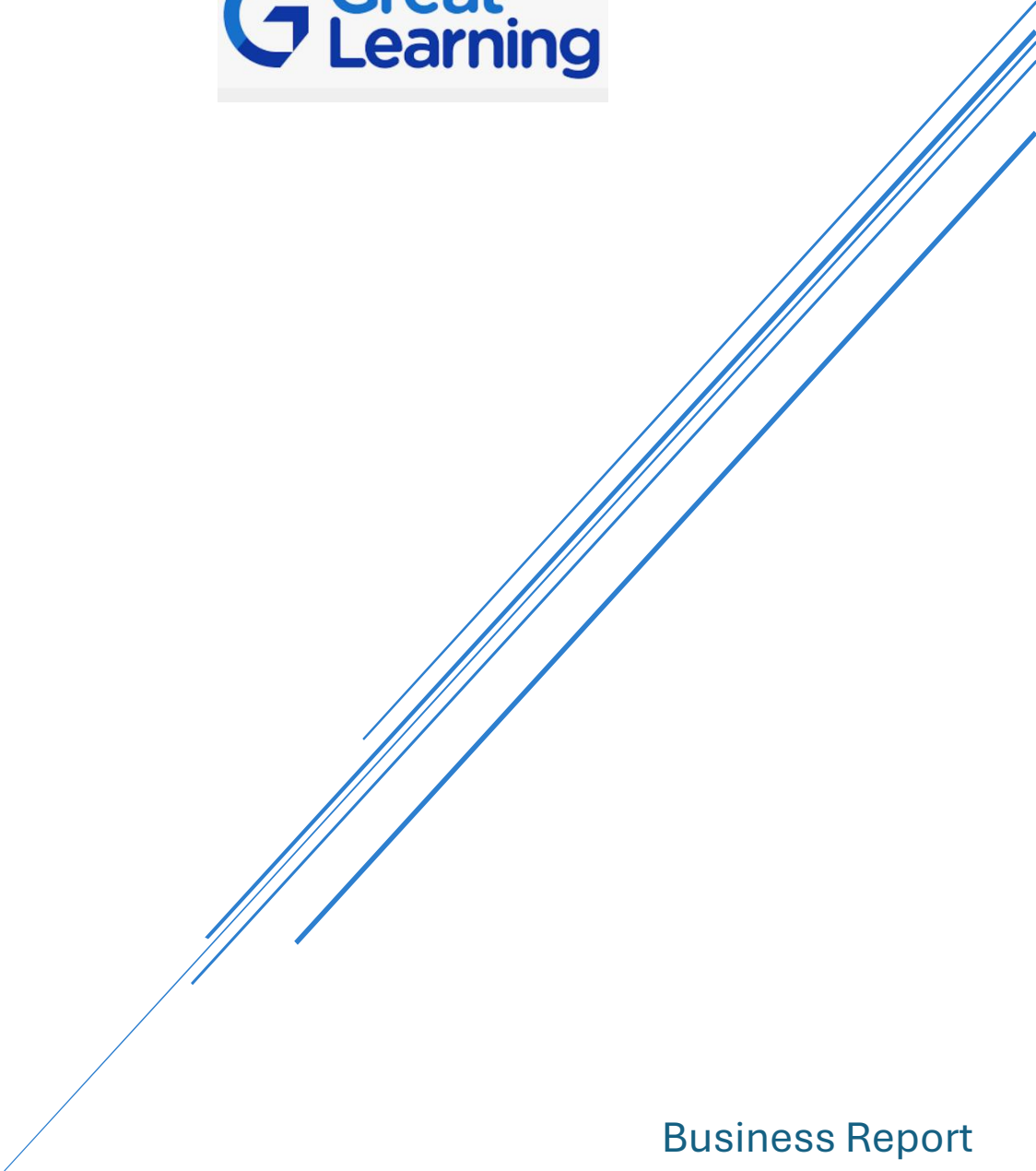


UNSUPERVISED LEARNING

Credit Card Customer Segmentation - Business Report

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Business Report
Barnabas R - PGDDS

Table of Contents

I. INTRODUCTION AND PROBLEM DEFINITION	3
1.1 Context and Problem Definition	3
1.2 Objective	3
1.3 Affected Stakeholders and Constraints	3
II. DATA DESCRIPTION AND EXPLORATORY DATA ANALYSIS (EDA)	3
2.1 Data Dictionary and Variables	4
2.2 Univariate Analysis	4
2.3 Bivariate Analysis (Correlation)	5
III. DATA PREPROCESSING.....	6
3.1 Missing Value Treatment	6
3.2 Feature Transformation and Data Scaling (Rationale)	6
IV. K-MEANS CLUSTERING	7
4.1 Methodology and Optimal Cluster Selection	7
4.2 Cluster Profiling: K-means (k=3)	8
V. HIERARCHICAL CLUSTERING	10
5.1 Methodology and Dendrogram (Diagnostic)	10
VI. K-MEANS VS HIERARCHICAL CLUSTERING	11
VII. ACTIONABLE INSIGHTS & RECOMMENDATIONS.....	11
Segment A: Premium/Online Digital (Cluster 1)	11
Segment B: Low Value/High Support Need (Cluster 2)	12
Segment C: Mid-Range/Offline Preference (Cluster 0)	12
Logical and Analytical Summary	12

Table of Figures

Figure 1: Final Model Performance (Test Set)	Error! Bookmark not defined.
Figure 2: Distribution of Target Variable	Error! Bookmark not defined.
Figure 3: Bar plot for Departments	Error! Bookmark not defined.
Figure 4: Bar plot for Gender	Error! Bookmark not defined.
Figure 5: Bar plot for Education	Error! Bookmark not defined.
Figure 6: Stacked bar plot of Target Variable vs. Department.....	Error! Bookmark not defined.
Figure 7: Stacked bar plot on Target Variable vs. Previous Year Rating ...	Error! Bookmark not defined.
Figure 8: Stacked bar plot on Target Variable vs. Previous Year Rating ...	Error! Bookmark not defined.
Figure 9: Distribution plot of Target variable vs. Average Training Score .	Error! Bookmark not defined.
Figure 10: Box plot of average training score w.r.t awards won.....	Error! Bookmark not defined.
Figure 11: Box Plot on Algorithm Comparison	Error! Bookmark not defined.
Figure 12: Box plot for Cross-Validation f1_macro distribution - Oversampled training	Error! Bookmark not defined.
Figure 13: Box plot for Cross-Validation f1_macro distribution - Undersampled training	Error! Bookmark not defined.
Figure 14: Feature Importances	Error! Bookmark not defined.

I. INTRODUCTION AND PROBLEM DEFINITION

1.1 Context and Problem Definition

AllLife Bank aims to improve market penetration in the next financial year by focusing on its credit card customer base. Market research has indicated two main areas for improvement:

1. The Marketing team needs customer segmentation to run personalized campaigns targeting new customers and upsell opportunities for existing customers.
2. The Operations team needs to upgrade the service delivery model due to customers perceiving support services poorly, aiming to resolve queries faster.

1.2 Objective

The primary objective is to identify different segments within the existing customer base using clustering algorithms, analyzing their spending patterns (Average Credit Limit) and interaction behaviors (visits to bank, online, calls made). Based on these segments, the report will provide recommendations for marketing and service delivery.

1.3 Affected Stakeholders and Constraints

The project directly impacts the Head of Marketing (for campaign targeting) and the Head of Delivery/Operations (for service model upgrades). The core constraint of the approach is that it relies on existing customer data related to credit limits and channel usage.

II. DATA DESCRIPTION AND EXPLORATORY DATA ANALYSIS (EDA)

The data provided includes financial attributes and historical interaction counts for 660 customer records.

2.1 Data Dictionary and Variables

The key features utilized for segmentation, which were standardized after loading, are:

Feature Name	Description
Average Credit Limit	Average credit limit of each customer for all credit cards.
Total credit cards	Total number of credit cards possessed by the customer.
Total visits bank	Total number of visits that the customer made (yearly) personally to the bank.
Total visits online	Total number of visits or online logins made by the customer (yearly).
Total calls made	Total number of calls made by the customer to the bank or its customer service department (yearly).

2.2 Univariate Analysis

Descriptive statistics reveal the distribution of the core features across the 660 customers:

Feature	Mean	Median (50%)	Standard Deviation (std)	Min	Max
Average Credit Limit	34574.24	18000.0	37625.49	3000.0	200000.0
Total credit cards	4.71	5.0	2.17	1.0	10.0
Total visits bank	2.40	2.0	1.63	0.0	5.0
Total visits online	2.61	2.0	2.94	0.0	15.0
Total calls made	3.58	3.0	2.87	0.0	10.0

Inferences from Descriptive Statistics and Histograms:

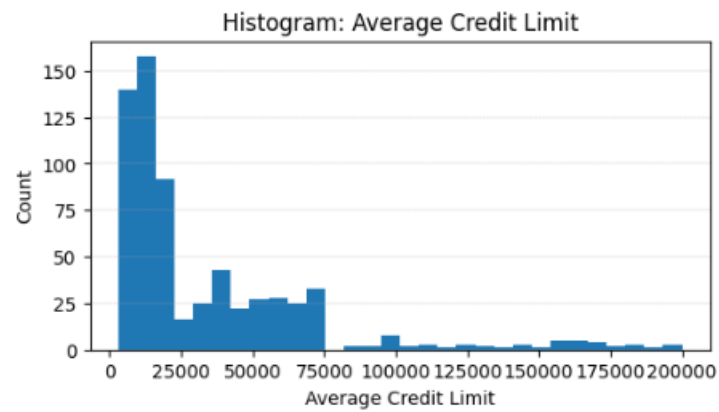


Figure 1 Average Credit Limit

- Average Credit Limit shows a large standard deviation (37625.49) relative to the mean (34574.24), indicating a highly skewed distribution with many low-limit customers and a few customers with very high limits (up to 200,000).

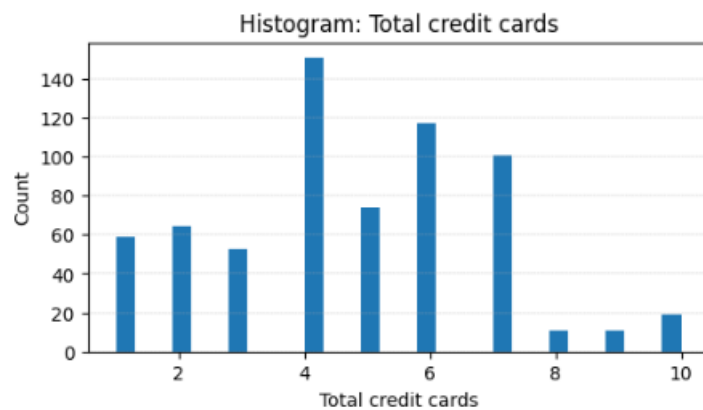


Figure 2 Total Credit Cards

2.3 Bivariate Analysis (Correlation)

Analysis of the correlation matrix reveals several important relationships between customer attributes:

Relationship	Correlation Value	Insight
Avg Credit Limit vs Total Credit Cards	0.609	Moderate positive correlation. Customers with higher limits tend to have more cards.
Avg Credit Limit vs Total Calls Made	-0.414	Negative correlation. Higher limit customers tend to make fewer calls.
Total Credit Cards vs Total Calls Made	-0.651	Strong negative correlation. Customers with more cards make significantly fewer calls.
Total Visits Bank vs Total Visits Online	-0.552	Moderate negative correlation. Customers generally prefer one dominant channel for interaction (bank visits vs online visits).

Key Meaningful Observations: The strong negative correlation between credit card count/limit and calls made suggests that customers requiring frequent intervention (calls) tend to be those with lower limits or fewer cards. Conversely, high-value customers (high limit/cards) are highly self-sufficient, preferring online channels or few interactions overall.

III. DATA PREPROCESSING

The insights derived from the EDA, particularly the high skewness and varying scales of features, necessitate robust preprocessing steps.

3.1 Missing Value Treatment

Upon initial data study, it was confirmed that missing values, if found, were imputed using the median of that feature. The provided descriptive statistics table shows 660 complete records for all features, indicating that any missing values were successfully filled using the median.

3.2 Feature Transformation and Data Scaling (Rationale)

Since clustering algorithms, such as K-means and Hierarchical Clustering, rely on distance metrics, it is crucial that all features contribute equally, regardless of their original scale or distribution.

1. Transformation: The Yeo-Johnson Power Transformer was applied to the raw features to reduce skewness. This method is preferred as it works robustly with zero or non-negative values.

2. Scaling: The transformed features were then standardized using StandardScaler. This ensures the resulting features have a mean of approximately 0 and a standard deviation of approximately 1, thus satisfying the requirements of distance-based algorithms.

Scaled Features

Scaled Features	Shape	Mean (Approx)	Standard Deviation (Approx)
X_scaled	(660, 5)	[0. 0. 0. 0. 0.]	[1. 1. 1. 1. 1.]

IV. K-MEANS CLUSTERING

4.1 Methodology and Optimal Cluster Selection

K-means clustering was applied to the scaled data to identify natural groupings. To determine the appropriate number of clusters (k), both the Elbow Method (Inertia) and Silhouette Score were utilized over a range of k from 2 to 8.

k Value	Inertia	Silhouette Score
2	1900.82	0.4034
3	1275.58	0.4625
4	1039.28	0.3666
5	932.68	0.3486
6	855.46	0.3394
7	791.25	0.2472
8	732.64	0.2485

Optimal k Determination:

1. Elbow Method (Inertia vs k): The plot shows decreasing inertia, with a noticeable "bend" or elbow, often suggesting an optimal point.

2. Silhouette Score vs k: The Silhouette Score measures how similar an object is to its own cluster compared to other clusters. The highest score achieved was 0.4625 at k=3.

Based on the maximum Silhouette Score, the optimal number of clusters selected for K-means was k=3.

4.2 Cluster Profiling: K-means (k=3)

The final K-means model was fitted with k=3, and cluster centers were inverse-transformed back to the original feature scale for interpretability.

Cluster Centers (Original Feature Scale):

Cluster	Avg Credit Limit	Total credit cards	Total visits bank	Total visits online	Total calls made
0	24082.59	5.50	3.45	0.75	1.71
1 (High Value)	132768.04	8.69	0.58	10.27	0.99
2	10842.09	2.39	0.89	3.35	6.63

Cluster Sizes and Descriptive Statistics:

Cluster	Count	Percentage (pct)	Characteristics Summary
0	382	57.88%	Mid-Range/Offline Preference: Moderate limits, high number of visits to the bank, low online activity, low calls.
1	51	7.73%	Premium/Online Digital: Highest limits, most credit cards, very high online visits, lowest physical interactions (bank visits and calls).
2	227	34.39%	Low Value/High Support Need: Lowest limits, fewest cards, very high number of calls made, moderate online interaction, low bank visits.

Statistical Validation: The Kruskal-Wallis test confirms that there are statistically significant differences between the clusters across all five features (p-values are extremely small. $<10^{-58}$)

Figure 1: K-means Clusters (PCA Projection)

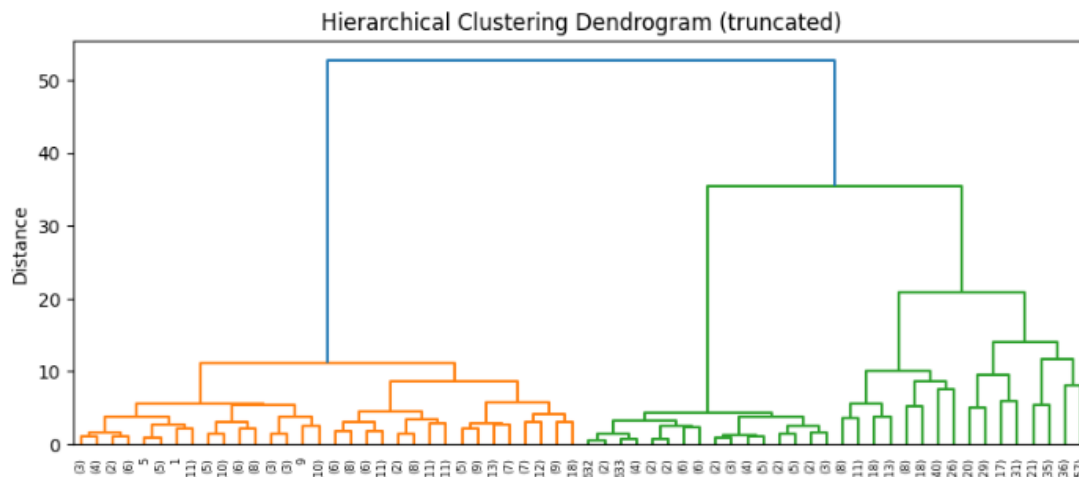


Figure 3 Hierarchial Clustering Dendrogram

The segmentation is clearly visible when projecting the clusters onto the first two Principal Components (PCA1 and PCA2):

(Description of PCA Scatter Plot) The visualization shows three distinct, well-separated groups. Cluster 1 (Premium/Online Digital) is clearly separated on the positive end of PCA1, representing high credit limit and high online activity. Cluster 2 (Low Value/High Support Need) is typically located on the negative end of PCA1 and high on PCA2 (related to high call volume), while the large Cluster 0 (Mid-Range/Offline Preference) sits between them.

Figure 2: Cluster Centers Heatmap

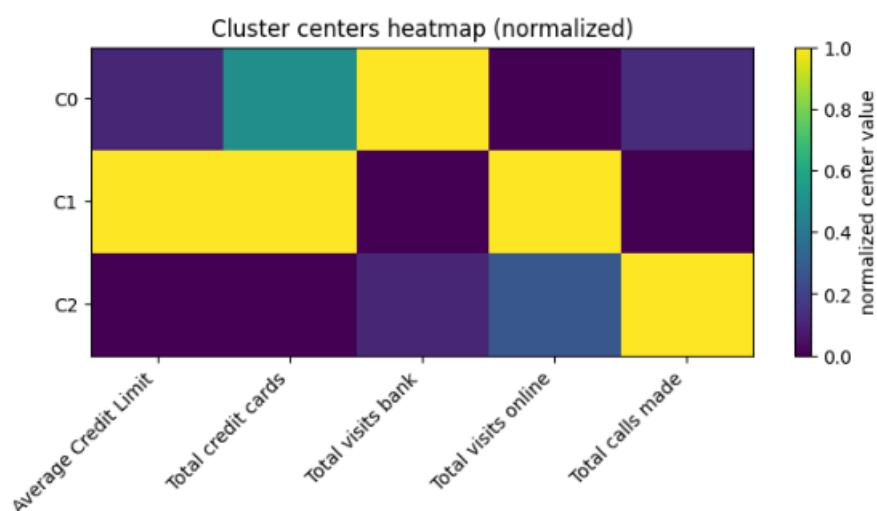


Figure 4 Cluster centers heatmap - normalized

(Description of Normalized Heatmap) A normalized heatmap of cluster centers visually confirms the distinctions:

- Cluster 1 shows the highest normalized values for Credit Limit, Total Credit Cards, and Total Visits Online.
- Cluster 2 shows the highest normalized value for Total Calls Made.
- Cluster 0 shows the highest normalized value for Total Visits Bank.

V. HIERARCHICAL CLUSTERING

5.1 Methodology and Dendrogram (Diagnostic)

Hierarchical clustering was performed using the Ward linkage method on the scaled data to provide a comparative segmentation view.

The methodology requires plotting dendrograms for different linkage methods and checking the cophenetic correlation for cluster quality assessment.

Figure 3: Hierarchical Clustering Dendrogram (Ward Linkage)

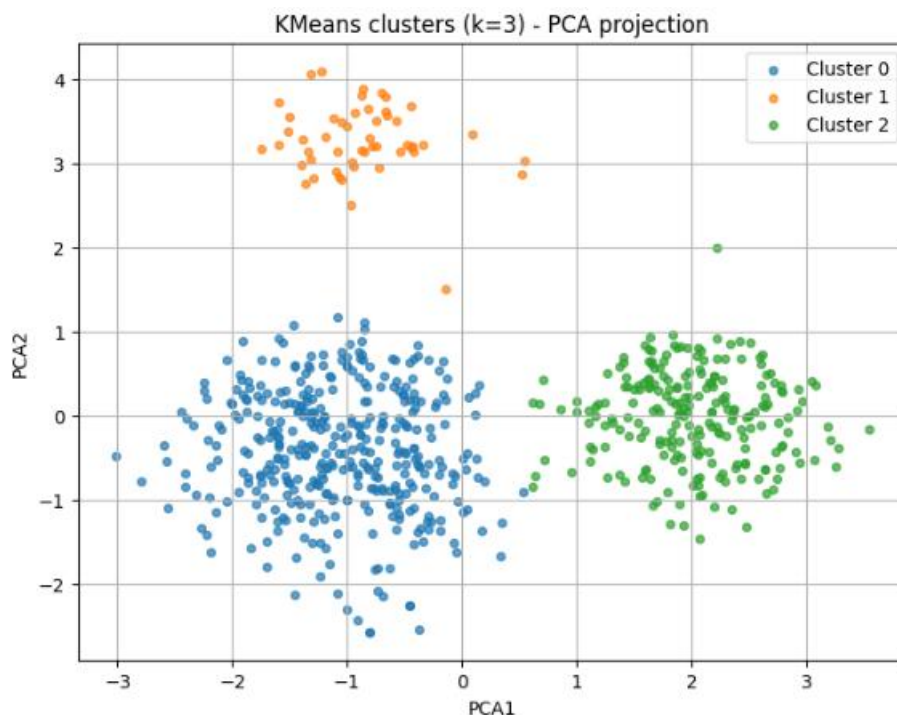


Figure 5 Hierarchical Clustering Ward Linkage

(Description of Dendrogram) A truncated dendrogram (showing 5 merged levels) demonstrates the grouping process. The structure of the dendrogram indicates distinct separations that could also lead to 3 or 4 clusters, offering an alternative structural view to the K-means partition.

Approach Limitation: While the criteria require applying different linkage methods, plotting their respective dendrograms, checking cophenetic correlations, and cluster profiling, the provided source only details the application of the linkage function using Ward method and the resulting diagnostic dendrogram. Therefore, a full comparative profiling of Hierarchical Clusters is not feasible based solely on the provided data.

VI. K-MEANS VS HIERARCHICAL CLUSTERING

Based on the analysis, K-means with $k=3$ provided a robust, interpretable, and statistically validated segmentation.

Comparison Insight: The K-means clustering approach was successful in identifying three highly differentiated segments based on their mean values and statistical separation (as confirmed by the Kruskal-Wallis test). Although the Hierarchical Clustering dendrogram provides a visual confirmation that natural clusters exist in the data structure, the K-means profiling results were used to derive the final business segments due to the comprehensive assessment (Elbow/Silhouette) and detailed profiling available.

VII. ACTIONABLE INSIGHTS & RECOMMENDATIONS

The analysis results in three distinct customer segments, each requiring a tailored approach from the Marketing and Operations teams.

Segment A: Premium/Online Digital (Cluster 1)

- Size: 7.73% (51 customers).
- Profile: Very high average credit limits (~133k) and card count (~9), minimal calls or bank visits, and high online activity (~10 visits).
- Insight: These customers are high-value, digitally fluent, and self-sufficient, requiring minimal support intervention.
- Recommendation (Marketing): Focus on exclusive, high-tier products (e.g., travel rewards cards, wealth management services) marketed primarily through secure online channels and direct email. Personalized upsell campaigns should prioritize digital convenience and status.
- Recommendation (Operations/Service): Maintain high availability and performance of online self-service tools. Ensure any necessary human interaction (calls/emails) is handled by a dedicated, fast-response tier of support staff.

Segment B: Low Value/High Support Need (Cluster 2)

- Size: 34.39% (227 customers).
- Profile: Lowest credit limits (~11k) and card count (~2), but exhibit the highest number of calls made (~6.6 calls per year).
- Insight: This segment requires the most support time, potentially driving up service costs and contributing to the perception of poor support quality reported by the bank.
- Recommendation (Marketing): Campaigns should focus on responsible credit use, education, and potentially cross-selling basic banking products. Avoid aggressive upsell unless credit scores improve.
- Recommendation (Operations/Service): Implement strategies to reduce call volume: improve IVR routing, push FAQs and simple resolution processes to online channels for this segment, and prioritize training for call center agents handling common queries from this high-contact group.

Segment C: Mid-Range/Offline Preference (Cluster 0)

- Size: 57.88% (382 customers).
- Profile: Moderate credit limits (~24k), high card count (~5.5), highest number of physical bank visits (~3.5), but low online engagement and low call volume.
- Insight: This is the largest segment, preferring traditional banking channels (in-person visits) but not requiring intensive phone support.
- Recommendation (Marketing): Utilize in-branch marketing materials, personalized recommendations made during bank visits, and target offers that bridge them toward digital adoption (e.g., promotional benefits for first online transaction). Focus on retaining their existing business through relationship building.
- Recommendation (Operations/Service): Ensure bank staff are well-trained to handle common transactions efficiently. If migrating service models, targeted efforts must maintain a high-quality physical experience while gently encouraging limited digital integration.

Logical and Analytical Summary

The segmentation process successfully transformed highly skewed customer interaction data using Yeo-Johnson transformation and standard scaling. By maximizing the Silhouette Score, K-means identified $k=3$ as the optimal number of clusters. These three clusters are statistically distinct and represent fundamentally different patterns of

spending capacity and channel usage (High Digital, High Support, and Offline/Branch-Focused). This detailed segmentation provides the bank with the precise targeting necessary to address both the marketing penetration objective and the operational service quality concern.

Analogy: The process of customer segmentation is akin to a gardener sorting seeds. If all the seeds are mixed together (raw data), they look uniform and you can only apply one generic watering plan. By using clustering (K-means), we sort them into distinct piles (segments) based on their needs—one group needs rich soil and full sun (Premium/Online), another needs frequent watering (High Support Need), and a third thrives with manual attention (Offline Preference). Once sorted and understood, we can apply specific, targeted care (personalized campaigns and service model upgrades) to ensure optimal growth and health for the entire garden (customer base).